BRCA1-methylated sporadic breast cancers are BRCA-like in showing a basal phenotype and absence of ER expression.

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BRCA1 mutations have been associated with hereditary breast cancer only. Recent studies indicate that a subgroup of sporadic breast cancer might also be associated with reduction in BRCA1 mRNA levels and protein expression. However, the mechanism of reduced mRNA and protein expression is yet not fully elucidated. This study aims to assess BRCA1 protein expression and the role of BRCA1 promoter methylation in sporadic breast cancer in North Indian population and to correlate these with known prognostic factors and molecular profiles of breast cancer. BRCA1 protein expression was normal (>50 % tumour cells) in 41 (43 %) cases, reduced (20-50 % tumour cells) in 33 (35 %) cases and absent/markedly reduced (<20 % tumour cells) in 21 (22.1 %) cases. Cases which were negative for BRCA1 protein were more frequently positive for basal markers (29 versus 5 %) and were more often ER-negative (62 versus 39 %) than BRCA1-positive tumours. Methylation of BRCA1 promoter region was seen in 11/45 cases (24 %). All 11 cases showing BRCA1 methylation had absent (eight cases) or reduced (three cases) BRCA1 protein expression. BRCA1 protein-negative tumours were more frequently basal marker-positive and ER-negative, highlighting the 'BRCAness' of sporadic breast cancer with loss of BRCA1 protein expression through promoter hypermethylation similar to hereditary breast cancer with BRCA1 mutations. Loss of BRCA1 in sporadic breast cancer suggests that therapeutics targeting BRCA1 pathway in hereditary breast cancer like PARP inhibitors might be used as therapeutic targets for sporadic breast tumours.